

SHWET PRAKASH

(Masters in Computer Science with 6 years of professional experience in Machine Learning)

shwet.prakash97@gmail.com | [Github](#) | [LinkedIn](#) | Contact - +91-9566218457

TECHNICAL SKILLS

Programming & frameworks - Python, TypeScript, SQL, PyTorch, Scikit-learn, Pydantic, Asyncio, Pandas, NumPy

LLM & Generative AI - LangChain, LangGraph, Multi-Agent Systems, RAG (Hybrid/Graph), OpenAI (GPT-4o/5.1), LiveKit (Voice AI), Hugging Face, Vector Databases (Qdrant, Pinecone, Chroma, FAISS), Prompt Engineering, Function Calling/Tool Use

Machine Learning - Regression, Classification, Tree-Based Models, Bagging, Boosting, Ensembles, Clustering, Dimensionality Reduction

Deep Learning & NLP - Transformers (BERT, RoBERTa), PEFT/LoRA Fine-tuning, Knowledge Distillation, Attention Mechanisms, RNN (LSTM/GRU), Word Embeddings, SetFit

ML Deployment & MLOps - Docker, MLflow, DVC, CML, GitHub Actions, Amazon SageMaker, FastAPI, Flask, Model Quantization

Cloud, CI Tools & Backend- AWS (EC2, ECS/Fargate, Lambda, S3, SES, API Gateway), Azure, NestJS, BullMQ, Redis, PostgreSQL, Neo4j, PrismaORM, CI/CD Pipelines

PROFESSIONAL EXPERIENCE

Hexaware, Bangalore India - Full-time

Feb 2026 – Present

Designation: Senior Machine Learning Engineer

Technologies: Python, FastAPI, Azure, LangGraph, LangChain, Langfuse, RAG, PostgreSQL, Asyncpg, Redis, Neo4j, WebSockets, Docker, Pydantic, OAuth2, JWT Authentication, Uvicorn, OpenAI

- o Architected and deployed **Seagate Agent Assistant**, a production-grade multi-agent AI system using **LangGraph stategraph** and **FastAPI** to autonomously orchestrate and resolve **complex Seagate ITOps workflows** on Azure.
- o Engineered a highly-concurrent, asynchronous backend (**Asyncpg, Redis**) featuring **FastAPI WebSockets** and **custom LangGraph astream events** to stream LLM token generation and real-time micro-progress indicators directly to the web UI.
- o Implemented robust, persistent conversational memory by integrating **LangGraph's AsyncPostgresSaver** for transactional thread checkpointing and **AsyncRedisStore** for cross-session global state tracking.
- o Scaled and containerized the full-stack AI infrastructure via **Docker**, establishing **managed PostgreSQL connection pools**, stateless dual-channel **OAuth2 JWT authentication**
- o Architected **RAG pipeline** featuring sequential ingestion engine with **dual recursive/semantic chunking strategies**; integrated **Milvus (Zilliz)** for high-dimensional vector retrieval, **history-aware query rewriting** for multi-turn conversation, and an **LLM re-ranking layer** that significantly improved context relevance before generation.
- o Implemented a **Redis semantic answer cache** to optimize response latency for repeated queries and developed an **LLM-as-a-judge evaluation framework** to automate scoring for response **accuracy, completeness, and relevance**; integrated **real-time cost telemetry** for granular analysis of token expenditures during ingestion and embedding stages.

Quant.ai, Bangalore India - Full-time

June 2025 – Jan 2026

Designation: Senior Machine Learning Engineer

Technologies: Python, FastAPI, LangGraph, LangChain, Multi-Agent Systems, OpenAI (GPT-5.1), LiveKit, Hugging Face Transformers, Hybrid RAG, Qdrant, PostgreSQL, Asyncio, MLflow, Docker, GitHub Actions, AWS (EC2, ECS, ECR), Pydantic

- o Architected a **multi-agent conversational AI** for **airline booking** leveraging **LangGraph Swarm, FastAPI streaming, and PostgreSQL persistence**, implementing 4 specialized agents with async connection pooling and **session state management** to orchestrate stateful flight search, multi-city bookings, fare selection, change flight, check-in, and AI-fueled loyalty workflows, deployed via Docker Compose with 75-80% latency reduction through **parallel async execution**.
- o Architected **multi-agent voice AI system** on **LiveKit Cloud** with 2 specialized agents (Category Discovery Agent and Product Suggestion Agent), 15 GPT-5.1 function tools, **session-based state management**, real-time voice pipeline (**VAD→STT→LLM→TTS**) with interruption handling, and middleware API integration for dynamic furniture catalog querying (decision tree navigation, 50+ categories, series-based filtering, nested product customization).
- o Developed intelligent product search using **hybrid RAG** with Qdrant vector DB, OpenAI embeddings (text-embedding-3-small), LLM-based query decomposition (GPT-4o-mini), metadata filtering (price/category/dimensions) and **async FastAPI deployment** with Docker
- o Engineered a production-grade **MLOps pipeline** to automate the deployment of Hugging Face Transformer models (BERT, RoBERTa) on AWS. Leveraged **MLflow** for experiment tracking and model versioning, containerized a FastAPI-based API with Docker, and built a full CI/CD workflow with GitHub Actions to enable zero-downtime deployments to a scalable AWS ECS and ECR environment.

Ohai.ai, Kolkata, India - Full-time

Jan 2024 – May 2025

Designation: Senior Machine Learning Engineer

Technologies: Python, FastAPI, Typescript, Nestjs, Langchain, Openai, GPT-4o, LLM, Agents, RAG, Vector Database, Pinecone,

Multimodal AI, AWS (SES, S3, cloudfront, EC2), PrismaORM, Redis, BullMQ Pro, Docker, PostgreSQL, Knowledge graph

- o Designed and implemented a production-grade **LLM agent stack** using **LangChain**, tailored for a household assistant product with a **modular toolchain** of **25+ custom tools** (calendar, reminders, messaging, email, household management, meal planning etc.), each built with conditional logic, and a **dynamic system prompt** enriched with contextual chat history to drive accurate tool selection and intelligent agent behavior.
- o Implemented **Redis-based low-latency caching** to retain conversational context, optimizing AI assistant response times and minimizing database load in a real-time production environment.
- o Architected **scalable background processing** pipelines using **BullMQ Pro** for async email scanning and multimodal document parsing (**email/PDF/image** → **LLM-based extraction**); enabled job-level concurrency, fault-tolerant retries, smart rate limiting and downstream task orchestration
- o Built a robust LLM-based **document understanding engine** using **Openai GPT-4o**; with dynamic system prompt, **multimodal inputs (image, email, text)**, schema-constrained JSON output, and downstream post-processing to extract structured actions (titles, events, todos, reminders)
- o Implemented a **Neo4j-based knowledge graph** system to model and query **calendar events** using LLMgenerated **Cypher queries**, enabling semantic search over event metadata
- o Fine-tuned and benchmarked **open-source LLMs** using **LoRA** and Hugging Face for the task of classifying tools and their structured arguments
- o Integrated **cloud-native infrastructure** into a **NestJS application**, leveraging **AWS services** (S3, SES, CloudFront, Secrets Manager) to support secure file handling and email workflows, and utilized **Prisma ORM** across services for efficient database operations.

Skuad, Bangalore, India - Full-time

May 2020 - Dec 2023

Designation: Applied Data Scientist

Technologies: Python, Deep Learning, Machine Learning, NLP, MLOps, Generative AI, AWS (S3, ECR, Lambda, Fargate, EC2), Pytorch, Huggingface, Tensorflow, GPT-3, LLM, BERT, Langchain, Mlflow, scikit learn, SQL, Boto3, Docker, Github actions, pandas, numpy, FAISS, chroma

- o Employed **GPT-4 models** to annotate job functions based on their job title, finetuned **BERT/Roberta/Distibert** models on this annotated data and applied **post-training quantization** techniques, resulting in a **2x enhancement in model inference latency** on CPU instances while keeping 99.8% of the accuracy
- o Developed a **SetFit Model (Few-Shot Learning approach)** on a limited dataset for a email-classification problem, then compressed it with **Knowledge Distillation** which improved latency by 4x while maintaining an accuracy of 93%
- o Finetuned **LLM models like GPT-3** to extract timelines and convert them into appropriate date and time format along with the rule based regex system from the Schedule based emails with an **overall accuracy of 97%**
- o Leveraged **Base LLM models, like ChatGPT**, for automated email response generation using **dynamic few-shot examples** driven by email categorization
- o Designed and implemented a **scalable NLP API** using **FastAPI** and **Docker**, deploying it on **AWS Fargate** with the aid of AWS ECS and ECR
- o Designed **MLOPs pipeline** using MLflow, **AWS EC2 instances**, and **GitHub Actions** to implement robust retraining schedules and strategies, ensuring continuous model accuracy and relevance

EDUCATION

- o B. Tech and M. Tech (Dual Degree) in Computer Science from **Indian Institute of Information Technology, Chennai, (2015 - 2020)**, CGPA : 7.8
- o Coursework: Data Science, Machine Learning, Big Data, DBMS, Programming Languages, Object oriented programming

PROJECTS

- o **Corrective RAG Chatbot with LangGraph**: Built an intelligent RAG-based chatbot that uses a self-corrective pipeline to evaluate retrieval relevance and trigger web search if needed, improving factual accuracy and reducing hallucinations. Developed using LangGraph to orchestrate the workflow and integrate vector search with dynamic query reformulation and corrective actions. [Medium Link](#)
- o **Production-Grade MLOps with Automated CI/CD**: A fully automated CI/CD pipeline using GitHub Actions, MLflow, FastAPI, Docker, and AWS (EC2, S3, ECR, ECS). Includes real-time monitoring with Prometheus and Grafana for scalable, production-ready ML deployments. [Medium Link](#)
- o **Reflection Agent with LangGraph**: Engineered a cyclic Reflection Agent using LangGraph StateGraph and GPT-5.1 to iteratively critique and refine outputs via Tavily Search. Integrated Remote Postgres Memory for state persistence and FastAPI Streaming for real-time token delivery. [Github Link](#)